1 (a) (i) Relative isotopic mass = $12 \times \frac{\text{mass of one atom of the isotope}}{\text{mass of one atom of } {}^{12}C}$ / AW; [1]

(ii) $Ar = \frac{7}{10} \times 68 + \frac{3}{10} \times 63$; [1]

= 63.6 ; [1]

(b) (i) Energy change when 1 mole of solid ionic compound is formed from its gaseous ions; [1]

(ii) Ionic radius ; [1]

The smaller the ions the higher the charge density and hence the stronger the attracting force; [1]

Ionic charge; [1]

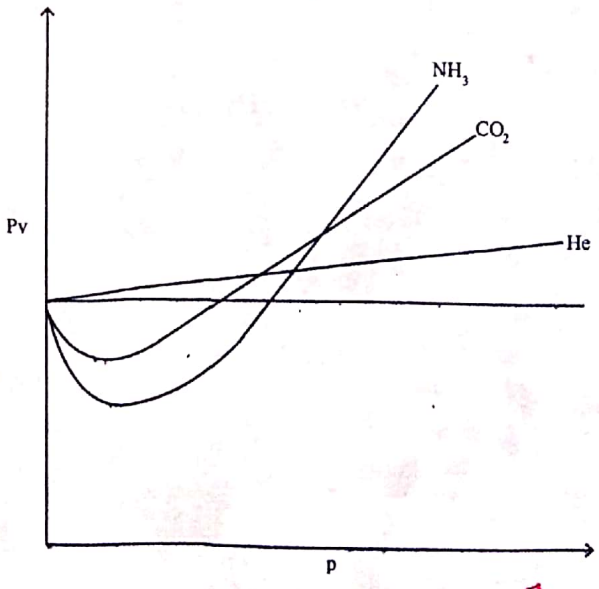
Ions with higher charge have higher charge density and hence strong force of attraction; [1]

(iii) $Mg^{2+}$ has a higher charge density than $Na^{+}$; [1]

- thus attract water molecules strongly giving a more negative $\Delta H_{hyd}$; [1]

(c) (i) - Highly electronegative atom (O, F, N) bonded to hydrogen; [1]
- At least a lone pair of electrons on highly electronegative atom; [1]

(ii)




[1]

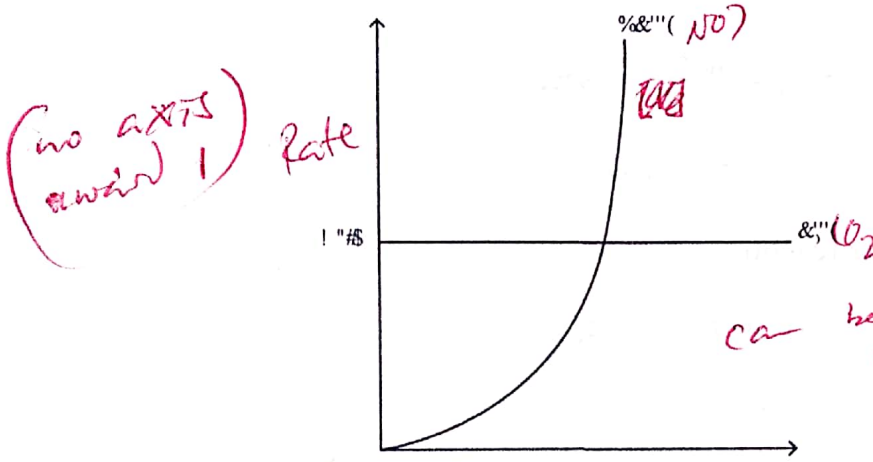
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(iii) - He has the weakest Van der Waal forces and slightly deviates from ideal behaviour; [1]
- Van der Waal forces in $CO_2$ have increased in strength thus increases deviation; [1]
- In $NH_3$ there are strong Hydrogen bonds thus has the greatest deviation; [1]

[Total : 15]

2 (a) (i) slowest step in a reaction mechanism; [1]

(ii)



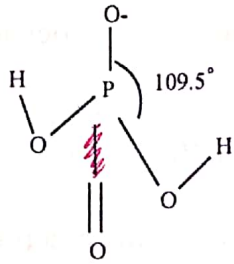

[1]

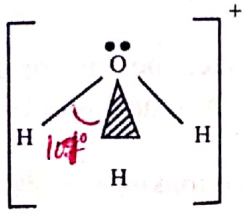
(iii) Y is higher temperature since reaction constant is higher; [1]

(b) (i) $H_3PO_4 + H_2O \rightarrow H_2PO_4^- + H_3O^+$; [1]

(ii) Acid base base acid; [1]

(iii) [1]




[1]

(c) (i) $Ag^+ + e^- \rightarrow Ag$

$n(Ag) = {}^{25}/_{108}$; [1]

$n(e^-) = n(Ag) = {}^{25}/_{108}$; [1]

$Q = nf = \frac{25}{108} \times 96500$; [1]

$= 22337.9C$

$t = \frac{Q}{I} = \frac{\frac{25}{108} \times 96500}{2}$ sec; [1]

$= 11\ 169$ sec

(ii) Anode reaction

$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$;

$V = \frac{24 \times \frac{25}{108} \times 96500}{4 \times 96500}$; [1]

$= 1.39dm^3$; [1]

(d) Anode potential = cathode potential
∴ ~~Number~~ flow of electricity; [1]
[Total: 15]

3 (a) When a system in equilibrium is subjected to a constraint, the system will adjust itself in a way which reverses the constraint and restore equilibrium (AW) [1]

(b) (i) 1 Forward reaction will be favoured since reaction is endothermic. [1]

~~- more water will dissociate into ions~~

2 Backward reaction will be favoured
Addition of $HCl_{(aq)}$ increases $[H^+]$ [1]

~~- hence background reaction is promoted to use up the added $H^+$ ions~~

(ii) $K_w = [OH^-_{(aq)}][H^+_{(aq)}]$ [1]

(iii) Concentration changes increase or decrease both the concentration of reactant and that of products proportionally hence no effect on $K_c$. [1]

Temperature change makes either forward or backward reaction to dominate leading to a different value of $K_c$. [1]

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(c) (i) $HIO_{3(aq)} \rightleftharpoons H^+_{(aq)} + IO^-_{3(aq)}$

| | | | |
|---|---|---|---|
| Initial | 0.50 | - | - |
| at eqm | (0.50 – 0.22) | 0.22 | 0.22 |
| | 0.28 | 0.22 | 0.2 |

[2]

(ii) $Ka = \frac{[H^+][IO_3^-]}{[HIO_3]}$ [1]

$= \frac{(0.22 moldm^{-3})^2}{0.28 moldm^{-3}}$ [1]

$= 0.173 moldm^{-3}$ [1]

[Total:15]

4 (a) (i) $CaCO_3 \rightarrow CaO + CO_2$ [1]

(ii) Calcium carbonate is an insulator in solid state;
hence the socket is safe to use. [2]

(iii) barium carbonate when digested is toxic; [1]

(b) (i) The reaction has high activation energy;
due to bond energy of Nitrogen triple bond. [2]

(ii) $Ca_3N_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2NH_3$ [1]

(iii) catalyst increases the rate of attainments of equilibrium; [1]

pressure of 200 atm is used because above it will increase. cost while below
reduce the yield of ammonia; [1]

moderate temperature (450 – 500)°C which if above will reduce the yield because it is exothermic but if below the rate of reaction is reduced. [1]

(c) $2NaHCO_{3(s)} \xrightarrow{less} Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$; [1]

decomposes when heated, [1]

$CO_{2(g)}$ and $H_2O_{(g)}$ are lost; [1]

causing a decrease in mass; [1]

sodium carbonate and sodium sulphate are stable; [1]

[Total:15]

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5 (a) (i) 1. $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$ [1]

decomposes;

purple fumes (of $I_{2(g)}$); [1]

2. $2HBr_{(g)} \rightarrow H_{2(g)} + Br_2(g)$; [1]

reddish/brown fumes of $Br_2(g)$; [1]

(ii) 1. reddish/brown colour of Br; [1]

$Cl_{2(g)} + 2Br^-_{(aq)} \rightarrow 2Cl^-_{(aq)} + Br_{2(g)}$; [1]

displacement reaction/

2. black solid of $I_2$; [1]

$Cl_2 + 2I^- \rightarrow 2Cl^- + I_{2(s)}$ [1]

$Cl_2$ is a powerful oxidise agent. [1]

(b) (i) two separate layers are formed; [1]

no reaction; insoluble in water; [1]

white fumes (of $HCl$) are observed; [1]

$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$ [1]

C in $Cl_4$ does not have 'd' orbital to accept have pair of electrons from water; [1]

(ii) has a giant molecular structure; [1]

inflammable;

reacts with $CaCo_3$ a base to form stronger $CaSiO_3$ when mixed with water; [1]

(c) (i) electrode in lead-acid battery [1]

(ii) $PbO_2 + 4H^+ + 2e^- \rightarrow Pb^{2+} + 2H_2O$ $E^\circ/V$ 1.47 [1]

$2C^- \rightarrow Cl_2 + 2e^-$ -1.36 [1]

(iii) $PbO_2 + 4H^+ + 2Cl^- \rightarrow Pb^{2+} + Cl_2 + 2H_2O$ [1]

$E^\theta = 1.47 - 1.36$

$= 0.11V$ [1]

[Total:15]

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[Total:15]

6 (a) (i) A reaction where one atom or a group of atoms are replaced by another; [1]

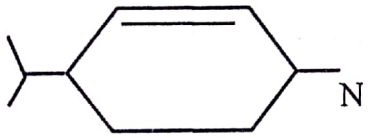
(ii) Volatility decreases as the number of carbon atoms increases. [1]

- Van der Waal forces increase in strength as molecular size increases. [1]

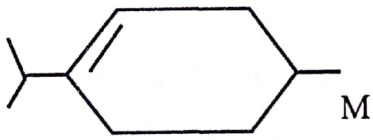
(iii)

| addition | condensation |
|---|---|
| no by product one product | small molecule removed as by product; [1] |
| usually similar monomers | usually 2 different monomers; [1] |

(b) (i) unsaturated saturated monomer




[1]




[1]

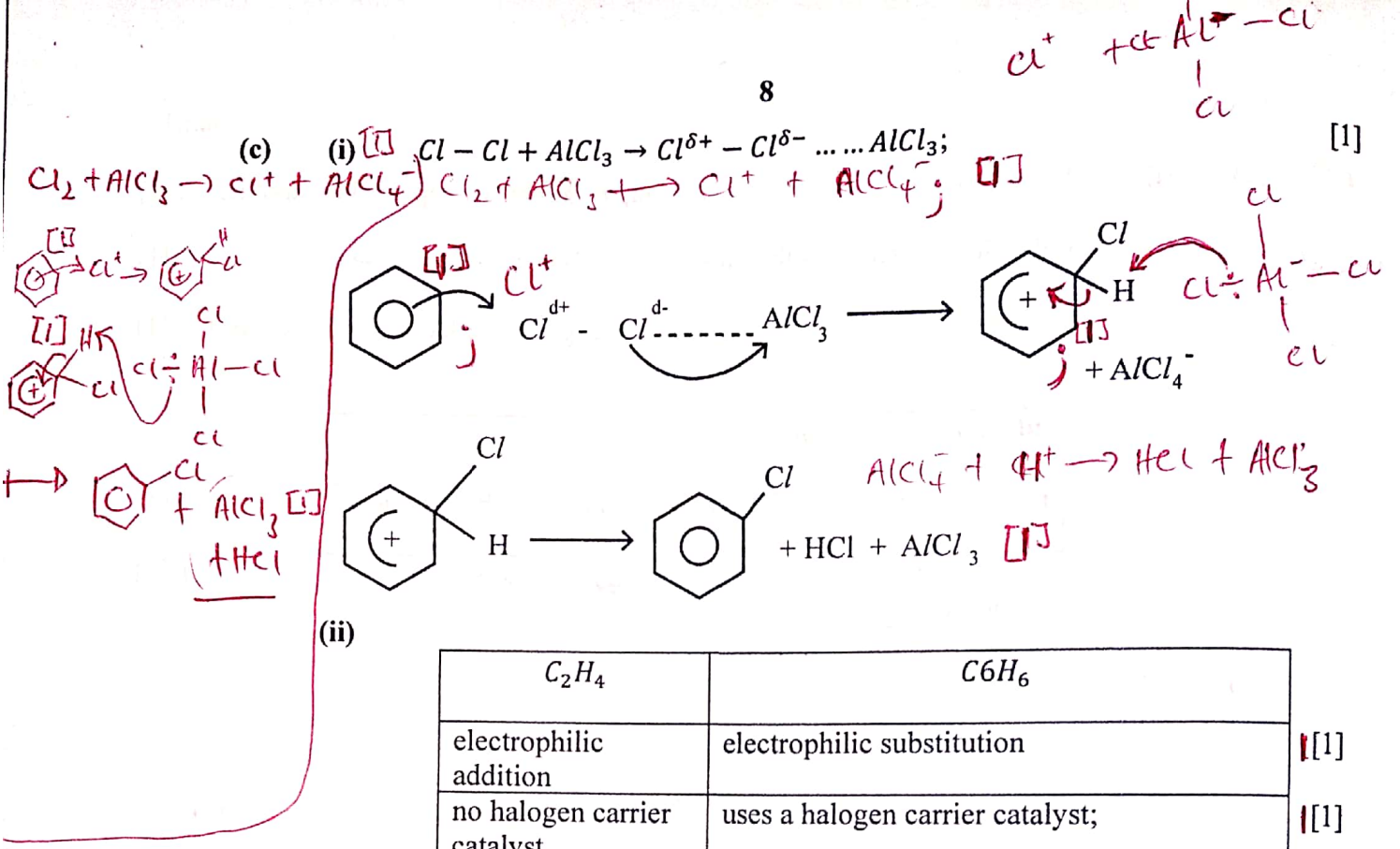
• $LiAlH_4$ reagent • dry ether condition or $NaBH_4$ reagent methanol condition [1] $H_2$ [1] Ni/Pt (120-150°

(ii) Conc $H_2SO_4$ / 170°C $Al_2O_3/SIO_2$ 450°C; [1]

[1]

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(c) (i) $Cl-Cl + AlCl_3 \rightarrow Cl^{\delta+}-Cl^{\delta-}\ldots\ldots AlCl_3$; [1]




(ii)

| $C_2H_4$ | $C6H_6$ | |
|---|---|---|
| electrophilic addition | electrophilic substitution | [1] |
| no halogen carrier catalyst | uses a halogen carrier catalyst; | [1] |

[Total:15]

7 (a) (i) Process in which shorter chain hydrocarbons are obtained from longer chain molecules/AW; [1]

(ii) $C_{16}H_{34} \rightarrow C_8H_{18} + 2C_2H_4 + C_4H_8 / C_{10}H_{22} + C_6H_{12}$; AW [1]

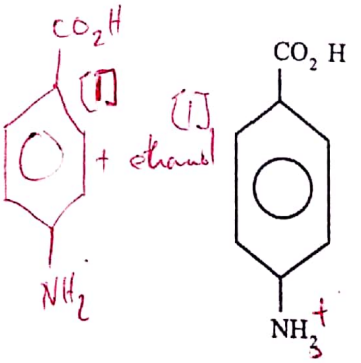
(iii) $Al_2O_3$ catalyst; / $SiO_2$ Alumina/Silica / Zeolite [1]
heat; high temp / 450°C ≥ high pressure (60–65 atm) [1]

(iv) fuels are produced; / more reactive alkenes are produced. [1]
cone

(b) (i) $HCl$, Sn/Fe; (reflux) [1]

$CH_3CH_2OH$ , conc $H_2SO_4$, reflux [1]

(ii) 1.



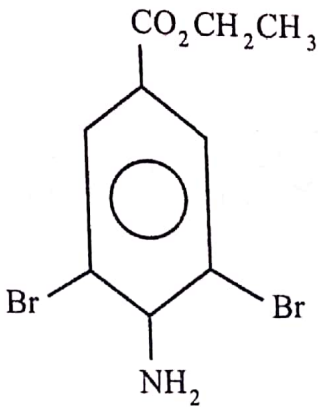

except a structures –

+ ethanol

$CH_3CH_2OH$

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**2.**

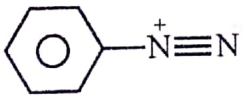



[1]

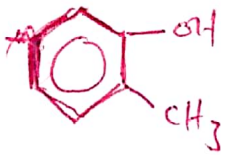
(iii) 4-nitrobenzoic acid is more soluble; [1]

- It can form more hydrogen bonds than B; [1]

**(c)** **(i)**




[1]




[1]

(ii) electrophilic sub
Coupling reaction; [1]

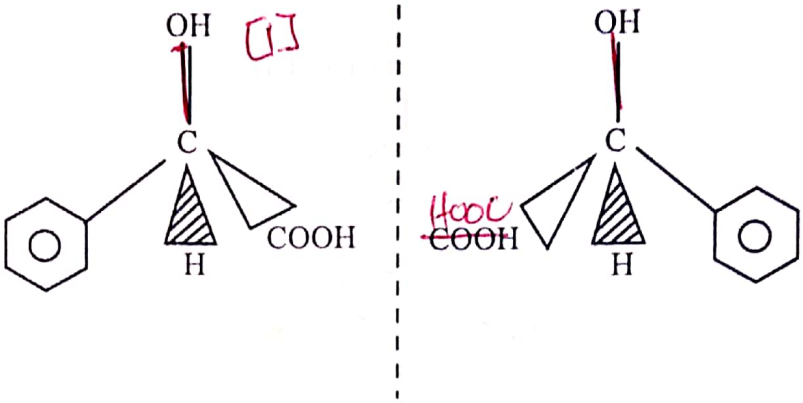
(iii) Slightly soluble due to bigger hydrophobic part; [1]
insoluble

[Total:15]

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8 (a) (i) Optical isomerism [1]

(ii)




[2]

(b) (iii) Hydrolysis [1]

(ii) 1. Reagents: HCN / NaCN [1]
Conditions: Presence of a trace base / $OH^-$ [1]

2. Reagents: dilute $H_2SO_4$ [1]
Conditions: Boil / Reflux (2) Heat [1]

(i) Increases the number of carbon atoms in a molecule / increases the length of the carbon chain in a molecule. [1]

(c) 1. Add 2.4 dinitrophenyl/hydrazine solution and a little acid and warm. [2]

Benzaldehyde gives a crystalline orange precipitate and no change occurs in Mandelic acid. [1]

OR

Add Tollen's reagent to a small amount of the sample, warm and leave to stand. Benzaldehyde produces a silver precipitate (silver mirror). No change with Mandelic acid. [1]

[Maximum 2 marks for 1 test]

2. Add sodium carbonate to the sample solution [1]
Mandelic acid produces bubbles of $CO_2$. [1]
No observation with benzaldehyde.

OR

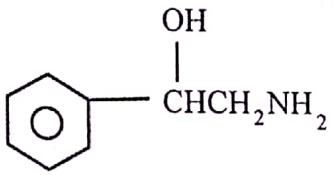
Add PCls to the sample solution. Add Na metal / [1]
no Observation for benzaldehyde [1]
white fumes on mandelic acid

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[Maximum 2 marks for 2 test]

**(d)** **(i)** ~~Reflux~~ with $LiAH_4$ / dry ether ® NaBH4 [1]

**(ii)**




[1]

**9** **(a)** **(i)** Engines designed to produce lower levels of harmful emissions by operating with excess air in the fuel-air mixture; AW [1]

**(ii)** Complete combustion hence lower levels of emissions; [1]

Catalytic converter – clean up the exhaust gases products of incomplete combustion react with oxides of nitrogen to produce less harmful gases ;

$2NO + 2CO \rightarrow N_2 + 2CO_2$ /

$C_xHy + \left(x + {}^{y}/_{4}\right) O_2 \rightarrow xCO_2 + {}^{y}/_{2}\, H_2O$ /

$C + O_2 \rightarrow CO_2$; filter pyr [1]

**(b)** **(i)** Sample mixture placed in a slot at the centre of gel buffered at a specific pH; [1]

current ®

Diagram accepted

- (High) voltage is applied across the gel; paper [1]
- ions separate by moving in response to the electrical field; [1]
- rate of migration of ion depend on charge density; /mass /size [1]

**(ii)** Paternity testing; [1]

any 3

- Identification of deceased; [1]
- Forensic investigation of crimes; [1]
- Establishing relationship between archaeological artefacts; [1]

gene mutation

Any 3

**(c)** Anode: $Fe_{(s)} \rightarrow Fe^{2+}_{aq} + 2e^-$; [1]

Cathode: $O_{2(g)} + 2H_2O(l) + 4e^- \rightarrow 4OH^-$; [1]

Overal: $Fe^{2+}_{aq} + 2OH^-_{aq} \rightarrow Fe(OH)_2$ ; [1]

Rusting

$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3$; [1]

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max 4

$2Fe(OH)_3 \rightarrow Fe_2O_3.XH_2O;$ [1]

(temperature)

[5 max 4]

10 (a) (i) 1. Type of solvent. [1]

2. Relative volumes of the two immiscible liquids. [1] [2]

(ii) 1. It must be insoluble in water. [1]

inert solvent

2. It must have high solubility for organic compounds. [1]

3. Must have a low bp for ease of evaporation. [1] max [2] [2]

volatile

(b) (i) 1. A buffered gel at a defined pH is set on the electrophoresis plate. [1]

2. A potential difference is set at the two ends of the plate. [1]

3. The mixture of amino acids I placed on the plate. Those which would have acquired a positive charge move to the cathode and those with a negative charge move to the positive terminal. The amino acids then separate out. [1]

4. Amino acids with larger charge moves faster. [1]

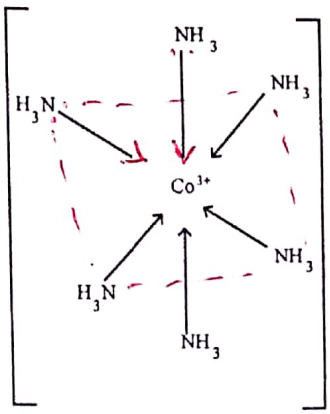
smaller mass/size/spherical shape affect speed

(ii) pH is an important factor since it determines the charge on the amino acid. [1] Different amino acids have different isoelectronic points hence at a given buffered pH they may have different net charge making it able to migrate to the electrodes. [1]

c) No of dative bonds around the central atom or ion [1]

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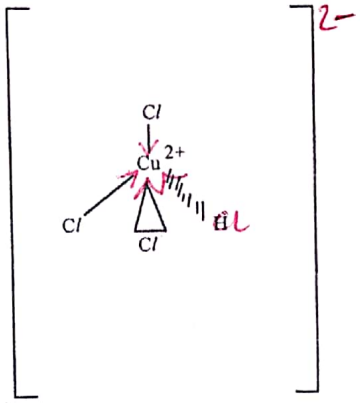
**(c)** **(i)**




; A

2

ignore arrows




2

- charge
- correct geometry

accept square planner

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